

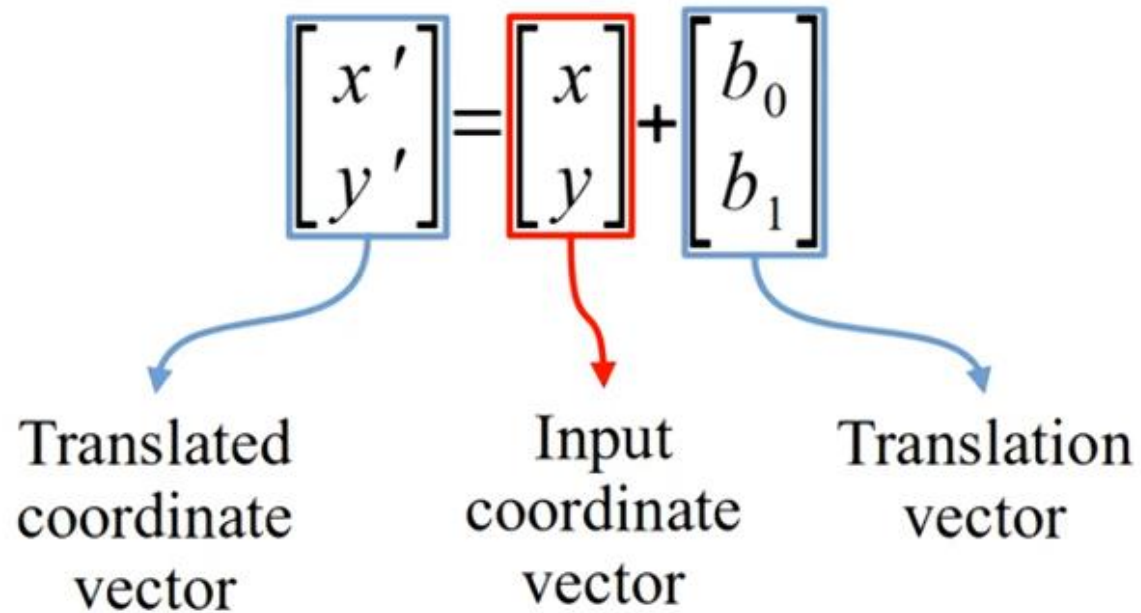
IMAGE TRANSLATION

Computer Vision with Hüseyin Özdemir

Translation equation in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

Translation equation in Cartesian coordinates



The diagram shows the translation equation in Cartesian coordinates: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$. The first vector is enclosed in a blue box, the second in a red box, and the third in a blue box. A blue arrow points from the first vector to the label "Translated coordinate vector". A red arrow points from the second vector to the label "Input coordinate vector". A blue arrow points from the third vector to the label "Translation vector".

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

Translated coordinate vector Input coordinate vector Translation vector

$$x' = x + b_0$$

$$y' = y + b_1$$

b_0 is translation in x direction

b_1 is translation in y direction

Translation equation in homogeneous coordinates

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & b_0 \\ 0 & 1 & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

In Cartesian coordinates

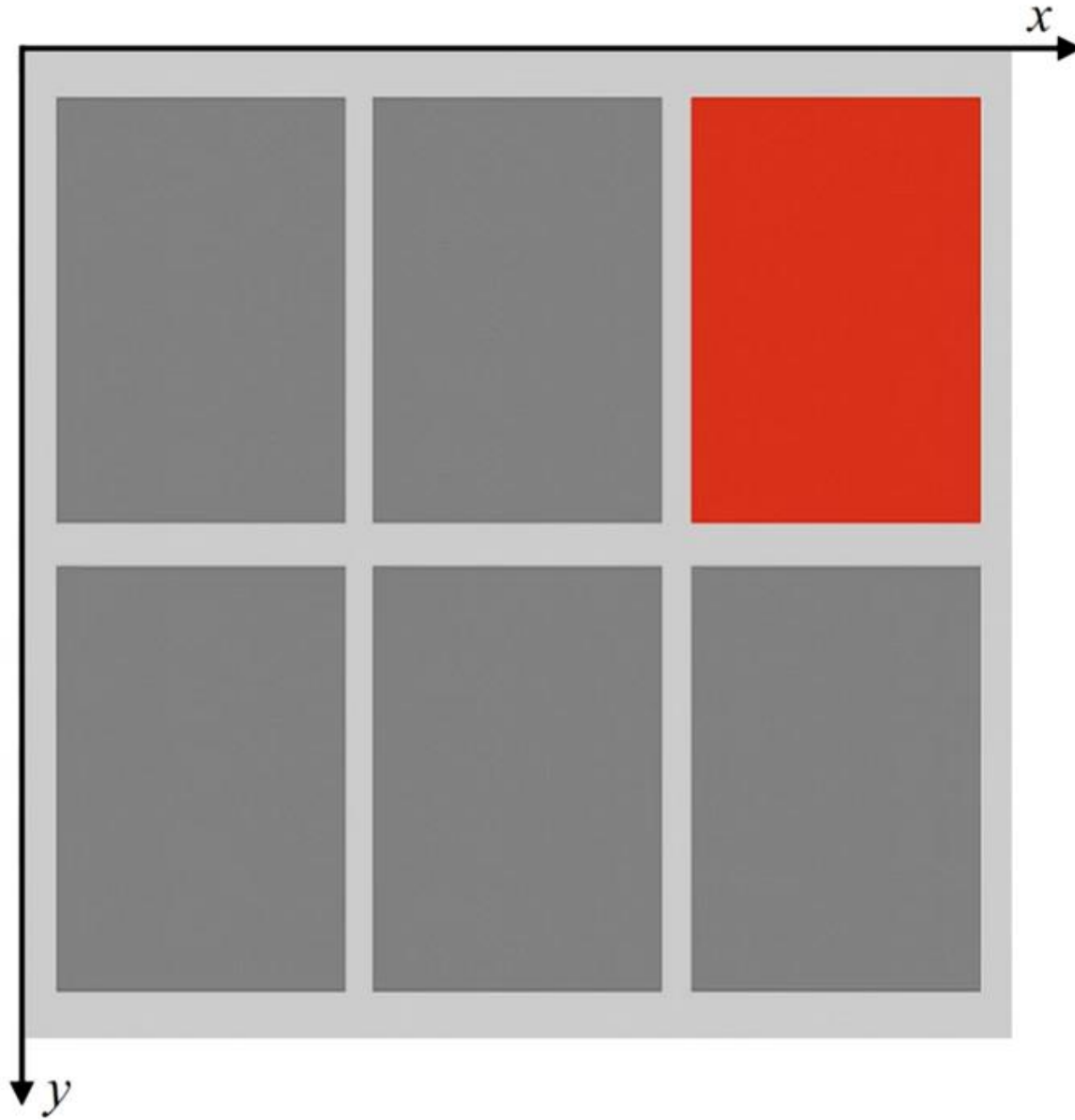
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

Vector addition

In homogeneous coordinates

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & b_0 \\ 0 & 1 & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

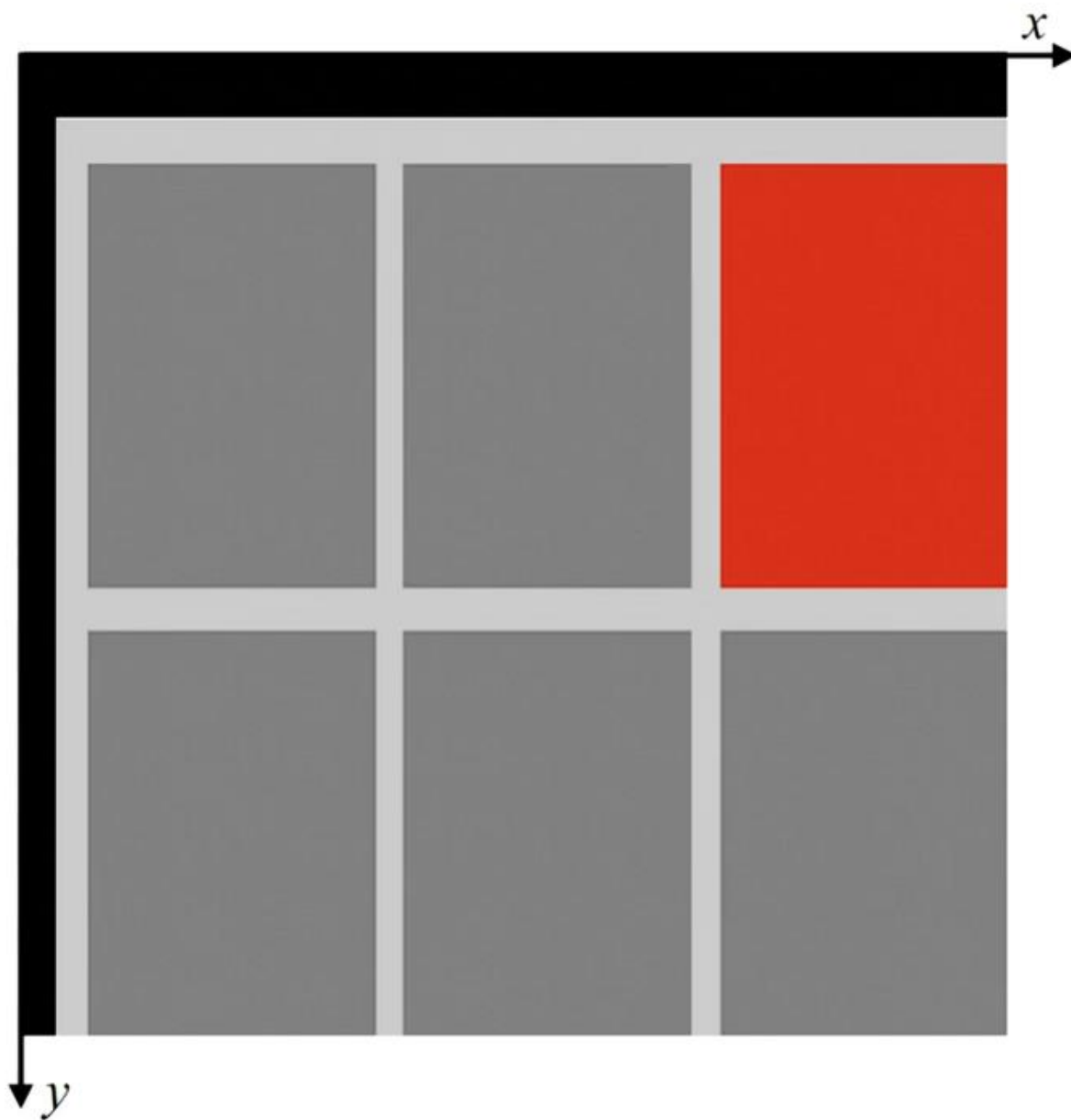
Matrix-vector multiplication



As an example, let horizontal and vertical translations be 30 and 60

Then, transformation matrix is

$$\begin{bmatrix} 1 & 0 & 30 \\ 0 & 1 & 60 \\ 0 & 0 & 1 \end{bmatrix}$$

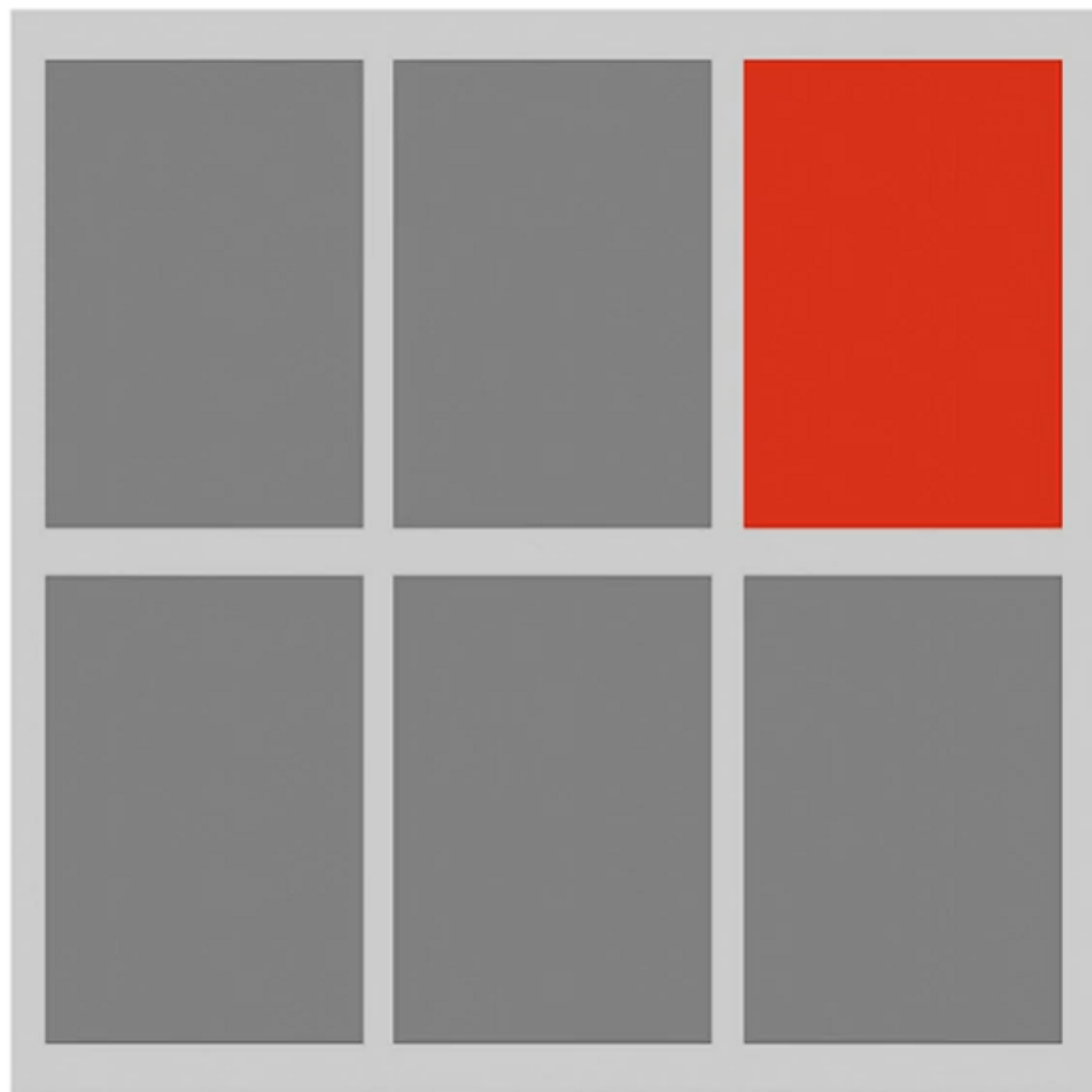


As an example, let horizontal and vertical translations be 30 and 60

Then, transformation matrix is

$$\begin{bmatrix} 1 & 0 & 30 \\ 0 & 1 & 60 \\ 0 & 0 & 1 \end{bmatrix}$$

$$b_0=30, b_1=60$$



Original Image



Translated Image

Reference

- <https://www.youtube.com/watch?v=EvhES3cQB8g>